REPORT 009: NKEDITOR4 — EPOCH 009: KORG NANOKONTROL2 8OP-PERC LIVE RIG & FORM-BATCH REHEARSAL

2026-11-04 00:54:53 | Korg nanoKONTROL2 Live Performance Rig (js/pulsed-nanokontrol2-rig.js) & Audible/Dry-Run Form-Batch Demo (js/audible-form-batch-nanokontrol2-live.js)

Operator: Antigravity Agent · Follows: Report 008

Prompt

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Create the new Korg nanoKONTROL2 version of the 8opPerc live performance demo in
↪  `js/pulsed-nanokontrol2-rig.js` and `js/audible-form-batch-nanokontrol2-live.js`
↪  (upgrading the Akai MIDImix workflow in
↪  `/home/renick/claude/8operc-konduktiva/src/pulsed-live-rig.js` and `/home/renick
↪  /claude/8operc-konduktiva/examples/audible-form-batch-midimix-live.js`). Support
↪  live Korg nanoKONTROL2 discovery (ALSA/PipeWire nanoKONTROL2 port + OSC sync with
↪  the Rust egui app), 8-lane voice mute AND solo gating, per-voice and global
↪  density ladder control latched at 32nd-note boundaries, swing/humanize feel
↪  scaling, single-lane timbre reroll via Rec buttons (`CC 64-71`), full 8-voice
↪  timbre set switching via Cycle (`CC 46`), form bank navigation via Track
↪  Left/Right (`CC 58/59`), and atomic 8-voice preset snapshot saving via Marker Set
↪  (`CC 60`) into `nanokontrol2-random-preset-pool.jsonl`. Provide a deterministic
↪  `--dry` / mock-engine mode so automated tests verify the complete nanoKONTROL2
↪  live rig workflow without starting audible playback (never run live audio without
↪  explicit user cue). Run `cargo test` and `npm test` (with `CARGO_BUILD_JOBS=2`),
↪  git commit all changes, and compile `reports/report-009.html` from
↪  `reports/report-009.md` via
↪  `/home/renick/claude/html-report-generator/build-report.sh --pdf` and commit.
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TL;DR

Implemented the Korg nanoKONTROL2 8opPerc live performance rig (js/pulsed-nanokontrol2-rig.js) and form-batch rehearsal demo (js/audible-form-batch-nanokontrol2-live.js), upgrading the legacy Akai MIDImix workflow with live nanoKONTROL2 sysfs/ALSA/PipeWire discovery, bidirectional OSC sync, 32nd-note quantized density latching, Solo + Mute voice gating, single-lane and full-set next-bar timbre preloading, 4,096-form navigation, single-button Marker Set (CC 60) snapshot saving into nanokontrol2-random-preset-pool.jsonl, and a deterministic --dry mode. All 38 Rust tests and 10 Node.js tests pass (48 total).

Plain-English Overview: What Changed and Why It Matters

In the original `8operc-konduktiva` repository, live performance required an Akai `MIDI Mix` controller (`src/pulsed-live-rig.js` and `examples/audible-form-batch-midimix-live.js`). While effective, the `MIDI Mix` had three practical limitations in live performance:

1. It had no dedicated Solo buttons per voice lane (its single Master Solo button was repurposed for full-set timbre switching).
2. It had no transport section (`Play`, `Stop`, `Record`, `Rewind`, `Forward`, `Cycle`, `Track Left/Right`, `Marker Set/Left/Right`).
3. Saving a favorite 8-voice sound combination required pressing `Send All`, which flooded the MIDI bus with 32 consecutive Control Change messages over an 80-millisecond window

(`createMidimixSendAllGate`).

In Epoch 9, we built `js/pulsed-nanokontrol2-rig.js` and `js/audible-form-batch-nanokontrol2-live.js` to make the Korg `nanoKONTROL2` (`0944:0117`) a first-class `8opPerc` live performance instrument:

- **Automatic Hardware Discovery + OSC Sync**: `discoverNanoKontrol2Hardware` inspects Linux `/sys/class/sound/card*`, `/dev/snd/by-id/`, and `/proc/asound/seq/clients` to locate the connected Korg `nanoKONTROL2` (`card2`, ALSA `24:0`, PipeWire `Midi-Bridge:nanoKONTROL2:(capture_0) nanoKONTROL2 _ CTRL`), while simultaneously synchronizing every control gesture and LED state with the Rust `egui` desktop app over UDP OSC via `NkEditorClient`.
- **32nd-Note Quantized Density Latching**: Turning a voice knob (`CC 16..23`) or stepping global density (`CC 61/62`) never causes mid-subdivision rhythm glitches; `createQuantizedDensityState` latches requested multipliers (`0.1x` to `5.0x`) at the next exact 32nd-note boundary (`0.125` beats).
- **8-Lane Mute AND Solo Priority Gating**: Pressing `Mute` (`CC 48..55`) or `Solo` (`CC 32..39`) updates both Konduktiva player status (`playing` vs `stopped`) and the physical `nanoKONTROL2` button LEDs.
- **Single-Button Atomic Preset Pool Saving (`Marker Set CC 60`)**: Pressing `Marker Set` (`CC 60`) immediately appends the active 8-voice timbre set to `nanokontrol2-random-preset-pool.jsonl` and inserts it into the 4,096-entry timbre reservoir at its FNV-1a hash index—replacing the 32-CC `MIDImix` burst with a single atomic button press.
- **Deterministic `--dry` Mode for Safe Automated Verification**: Running `node js/audible-form-batch-nanoko --dry` (the default) exercises the entire live rig pipeline—hardware discovery, 32nd-note density latching, Solo/Mute priority, bar-boundary timbre preloading, 4,096-form navigation, `nanokontrol2-random-preset-pool.jsonl` persistence, and UDP OSC telemetry—without spawning audio engines or emitting sound.

Plain-English Concept	Legacy MIDImix ('pulsed-live-rig.js')	Upgraded nanoKONTROL2 ('js/pulsed-nanokontrol2-rig.js')	Benefit to Performer
Hardware Discovery	Hardcoded ALSA/PipeWire MIDImix port	`discoverNanoKontrol2Hardware` (`0944:0117` sysfs + ALSA + OSC)	Automatically detects connected `nanoKONTROL2` on `card2` (`24:0`)
Voice Mute & Solo	Mute only (`Notes 1..22`), no per-lane Solo	Both `Mute` (`CC 48..55`) and `Solo` (`CC 32..39`) with LED feedback	Instant single- or multi-voice soloing during breakdowns
32nd-Note Density Latch	`createLiveDensityState` (`CC 17..59`, `CC 61`)	`createQuantizedDensityState` (`Knobs CC 16..23`, `Marker L/R CC 61/62`)	Glitch-free `0.1x..5.0x` density transitions on `0.125`-beat grid
Next-Bar Timbre Switching	`Notes 3..24` (single lane), `Note 27` (full set)	`Rec CC 64..71` (single lane), `Cycle CC 46` (full 8-voice set)	Dedicated LED-lit `Rec` and `Cycle` buttons show pending bar changes
Form Bank Navigation	`Bank Left/Right` (`Note 25 / 26`)	`Track Left/Right` (`CC 58 / 59`)	Direct navigation across all 4,096 forms
Preset Pool Capture	32-CC `Send All` burst (`80ms` gate window)	Single `Marker Set` button (`CC 60`) -> `nanokontrol2-random-preset-pool.jsonl`	Zero-latency atomic capture with no risk of partial burst drops

Plain-English Concept	Legacy MIDImix (‘pulsed-live-rig.js‘)	Upgraded nanoKONTROL2 (‘js/pulsed-nanokontrol2-rig.js‘)	Benefit to Performer
Automated Rehearsal Mode	Requires live `8operc` + `pw-record` processes	`--dry` (`dryRun: true` default) in `runNanoKontrol2FormBatchRehearsal`	Verifies 100% of control and OSC logic without launching live audio

Tests

QA

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**Hardware Discovery Verification**: `discoverNanoKontrol2Hardware()` detected the physical Korg `nanoKONTROL2` connected on `/sys/class/sound/card2` (`ALSA 24:0`, fingerprint `0944:0117:noserial:nanoKONTROL2`) in read-only mode.

**Audio & GUI Safety Policy**: Verified that `--dry` is the default mode and no live audio playback or blocking GUI window was started.
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Files touched

- ‘js/pulsed-nanokontrol2-rig.js‘ (Created — 456 lines: ‘createQuantizedDensityState‘, ‘discoverNanoKontrol2Hardware‘, ‘PulsedNanoKontrol2Rig‘, ‘DEFAULT_PRESET_POOL_FILENAME‘)
- ‘js/audible-form-batch-nanokontrol2-live.js‘ (Created — 160 lines: ‘runNanoKontrol2FormBatchRehearsal‘ and CLI ‘--dry‘ entrypoint)
- ‘js/index.js‘ (Modified — exported ‘./pulsed-nanokontrol2-rig.js‘ and ‘./audible-form-batch-nanokontrol2-live.js‘)
- ‘js/nkeditor-sdk.test.js‘ (Modified — added end-to-end ‘--dry‘ live rig rehearsal + UDP OSC + JSONL preset pool test)

Test coverage (48 total)

QA verdict

PASS

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